

Lists and functional programming

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List

We have seen vectors and matrices as data structures in R.

List is another important data structure in R.

It combines objects **with different types**.

Matrix/vector only supports a single type of data.

Similar to the struct in C programming.

See ART: Section 4.

Creating List

```
matteo_f <- list(name = "matteo",  
                 staff = TRUE,  
                 salary = 10)
```

This creates a list contains three elements:

- ▶ Character data: name
- ▶ Logical data: staff
- ▶ Numeric data: salary

name, staff and salary are called “tags” for values "matteo", TRUE, 10 respectively.

Displaying List

You can list all tags and their corresponding values by simply typing the name of the list on the console:

```
matteo_f
```

```
$name
```

```
[1] "matteo"
```

```
$staff
```

```
[1] TRUE
```

```
$salary
```

```
[1] 10
```

Indexing List

To obtain the value bound to a specific tag we can use the \$ sign:

```
matteo_f$staff
```

```
[1] TRUE
```

or similarly `matteo_f[["staff"]]`.

Or you can index a list without using tags:

```
matteo_f[[2]]
```

```
[1] TRUE
```

but note

```
matteo_f[2]
```

```
$staff
```

```
[1] TRUE
```

In particular note:

```
matteo_f[[3]] + 1
```

```
[1] 11
```

while

```
matteo_f[3]
```

```
$salary
```

```
[1] 10
```

```
class(matteo_f[3])
```

```
[1] "list"
```

So we can not do

```
matteo_f[3] + 1
```

```
# Error in matteo_f[3] + 1 : non-numeric argument  
# to binary operator
```

You can create a list with missing tags:

```
matteo_f <- list(name = "matteo", T, salary = 10)
```

Then you will have to access untagged elements using `matteo_f[[2]]`.

You can subset a list by doing:

```
matteo_f[c("name", "salary")]
```

```
$name
```

```
[1] "matteo"
```

```
$salary
```

```
[1] 10
```

or `matteo_f[c(1, 3)]`.

But you cannot do

```
matteo_f[[c(1, 3)]]
```

```
# Error in matteo_f[[1:3]] : subscript out of bounds
```

To add elements do

```
matteo_f <- list(name = "matteo", staff = T, sal = 10)
matteo_f$department <- "math"
matteo_f[[5]] <- 1985
matteo_f
```

\$name

[1] "matteo"

\$staff

[1] TRUE

\$sal

[1] 10

\$department

[1] "math"

[[5]]

[1] 1985

To remove an element do

```
matteo_f$department <- NULL  
matteo_f[[1]] <- NULL  
matteo_f
```

```
$staff  
[1] TRUE
```

```
$sal  
[1] 10
```

```
[[3]]  
[1] 1985
```

NOTE: before R 3.1.0 modifying a list triggered a copy of all elements, this is **not** true anymore! Modifying lists is much more efficient now.

To concatenate lists do:

```
more_info <- list(height = 182, employer = "UoB")  
matteo_f_2 <- c(matteo_f, more_info)  
matteo_f_2
```

\$staff

[1] TRUE

\$sal

[1] 10

[[3]]

[1] 1985

\$height

[1] 182

\$employer

[1] "UoB"

Note that this does something different:

```
( matteo_f_2 <- list("l1" = matteo_f, "l2" = more_info) )
```

```
$l1
```

```
$l1$staff
```

```
[1] TRUE
```

```
$l1$sal
```

```
[1] 10
```

```
$l1[[3]]
```

```
[1] 1985
```

```
$l2
```

```
$l2$height
```

```
[1] 182
```

```
$l2$employer
```

```
[1] "UoR"
```

To create a nested list from scratch do:

```
studs <- list(list(name = "Jack", age = 21),  
              list(name = "Tim", age = 20))  
studs
```

```
[[1]]
```

```
[[1]]$name
```

```
[1] "Jack"
```

```
[[1]]$age
```

```
[1] 21
```

```
[[2]]
```

```
[[2]]$name
```

```
[1] "Tim"
```

```
[[2]]$age
```

```
[1] 20
```

Then we can do:

```
course <- list(code = "MATH0017", year = "1",  
               students = studs)  
course[[1]]
```

```
[1] "MATH0017"
```

```
course[[3]][[2]]
```

```
$name
```

```
[1] "Tim"
```

```
$age
```

```
[1] 20
```

```
course$students[[2]]$name
```

```
[1] "Tim"
```

Functional Programming

So far, we have worked within a **Procedural Programming (PP)** paradigm.

In PP your program is divided into several subtasks and you write **functions** for each subtask.

Another common paradigm is Object Oriented Programming (OOP).

In OOP your program is divided into several pieces called “objects” and objects contain **data as well as procedures**.

PP and OOP divide the program **by features** thus is suitable for developing apps with complicated logics and components.

Functional Programming

However, most data science programs have a simple programming pipeline:

- ▶ Data1 -> Op1 -> Data2 -> Op2 -> ... -> Final Result
- ▶ Apply(Op1, Data1) -> Data2 -> Apply(Op2, Data2) -> Data3 -> ... -> Final Result

Functional Programming (FP) views our program as a pipeline, focusing on writing data-operating functions and applying such functions to our data.

R supports functional programming natively.

- ▶ C/C++ also supports functional programming via some advanced language features.
- ▶ Function pointers, templates, etc.

A Simple FP Example

Write a simple data operating function:

```
add <- function(x) { return( x + 1 ) }
```

Applying this function on some dummy data:

```
l <- list(1,2)  
lapply(l, add)
```

```
[[1]]
```

```
[1] 2
```

```
[[2]]
```

```
[1] 3
```

Here, `lapply` applies the `add` function to each element of the list `l`, producing a new list.

The input and output of `lapply` are both lists.

We can also convert the list output to a vector by using `unlist`:

```
l <- list(1,2)
unlist(lapply(l, add))
```

```
[1] 2 3
```

More conveniently, we can use `sapply`

```
l <- list(1,2)
sapply(l, add)
```

```
[1] 2 3
```

Looking at lapply more in detail:

```
args(lapply)
# function (X, FUN, ...)
```

the "..." (aka ellipsis) means that lapply accepts more arguments.

These will be passed to FUN. Example:

```
a_plus_b <- function(a, b){ a + b }
```

```
lapply(1:3, a_plus_b, b = 10)
```

```
[[1]]
[1] 11
```

```
[[2]]
[1] 12
```

```
[[3]]
[1] 13
```

The apply() function

Sometime we want to apply a function over rows or columns of a matrix.

For instance, consider the following matrix:

```
ns <- 3
marks <- matrix(c(78,85,56,80,79,88,34,90,51,76,84,47),
                 ncol=ns, byrow=TRUE)
marks
```

	[,1]	[,2]	[,3]
[1,]	78	85	56
[2,]	80	79	88
[3,]	34	90	51
[4,]	76	84	47

Assume that columns are students and rows are subjects.

To compute the maximum mark for each students we could do:

```
max_mark <- numeric(ns)
for(ii in 1:ns){
  max_mark[ii] <- max( marks[,ii] )
}
max_mark
```

```
[1] 80 90 88
```

We can do this more compactly with:

```
( max_mark <- apply(marks, 2, max) )
```

```
[1] 80 90 88
```

```
args(apply)
```

```
function (X, MARGIN, FUN, ..., simplify = TRUE)
NULL
```

To get the max mark per subject we do:

```
( max_mark <- apply(marks, 1, max) )
```

```
[1] 85 88 90 84
```

We can use any function that takes a vector in a returns a scalar:

```
apply(marks, 2, mean)
```

```
[1] 67.0 84.5 60.5
```

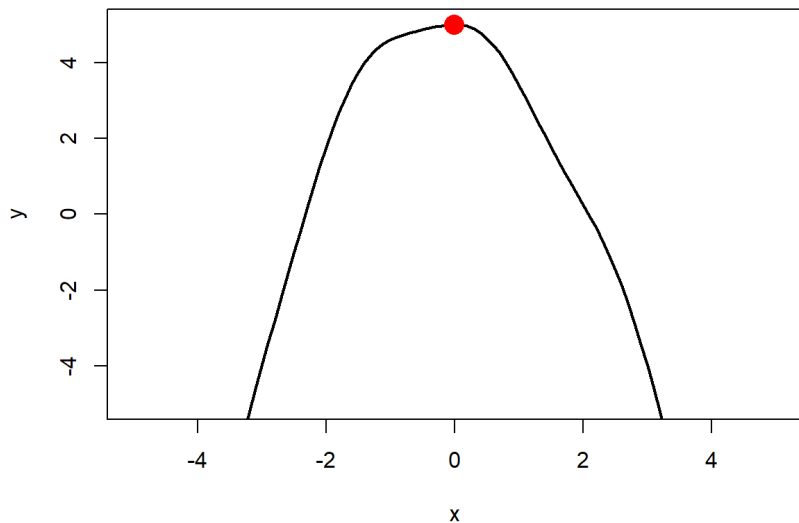
Functions are Variables

In FP, functions are variables too, **thus they can be passed to other functions as input arguments.**

In the previous example, `add` is a function that was passed to the `lapply` function as an input argument.

This property allows us to write clean and more readable code.

Gradient Ascent Revisited



```
f <- function (x){  
  return(-sin(x)^3-x^2+ 5)  
}
```

```
df <- function(x){  
  return(-3*sin(x)^2*cos(x) - 2*x)  
}
```

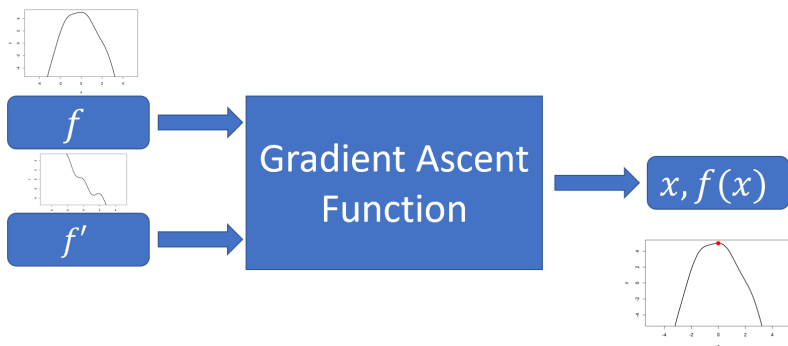
```
x <- -1.5  
while( abs(df(x)) > 0.01){  
  x <- x + 0.1 * df(x)  
}  
x
```

```
[1] -0.004884895
```

How can we wrap the gradient ascent algorithm using a function?

Gradient Ascent Function

Gradient ascent algorithm depends on f and df , thus this gradient ascent function should take **two functional** inputs.



Here is our function:

```
grad_asc <- function(x, f, df){  
  
  while( abs(df(x)) > .01){  
  
    x <- x + .1*df(x)  
  
  }  
  return(list(x, f(x)))  
}
```

```
grad_asc(x = -1.5, f, df)
```

```
[[1]]
```

```
[1] -0.004884895
```

```
[[2]]
```

```
[1] 4.999976
```

Gradient Ascent Function

In this example, the initial search point x is data and f and df are data operating functions.

`grad_asc` tells the program how f and df are applied to the data and produces the final outcome.

Functions are variables so we can put them in a list and simplify:

```
grad_asc <- function(problem){  
  f <- problem$func  
  df <- problem$deri  
  x <- problem$x  
  
  while( abs(df(x)) > .01){  
    x <- x + .1*df(x)  
  }  
  
  return(list(x, f(x)))  
}
```

```
problem <- list(x = -1.5, func = f, deri = df)  
  
unlist( grad_asc(problem) )
```

```
[1] -0.004884895  4.999976254
```

Conclusion

1. List in R can contain data with different types.
2. Lists can be nested or concatenated using `c()`.
3. It's important to be clear about `[[ ]]` vs `[ ]`.
4. FP focuses on data operating functions and how these functions are applied on data.
5. In FP, functions can be used as variables.